# Random Numbers in C++

PRNG properties + a tiny example

## PRNG PROPERTIES

Concepts

- **State size & speed:** memory footprint and performance.
- **Bits per output:** how many bits each random number has: 16 / 32 / 64 (sometimes odd sizes like 15).
- **Statistical quality:** how "random" the result is. All values should be equally likely; patterns (e.g. odd numbers appearing more often) indicate poor quality.
- **Period:** how long before the sequence repeats.
- **Predictability:** given some numbers generated by the PRNG, how easy can I predict the next one.

## GET A RANDOM NUMBER EXAMPLE

Code Example

Fixed seed → deterministic, reproducible results.

```
#include <random>

int main() {
  std::ranlux24_base rng(1234);
  std::uniform_int_distribution<int> dist(1, 6);

  int roll = dist(rng);
  return 0;
}
```

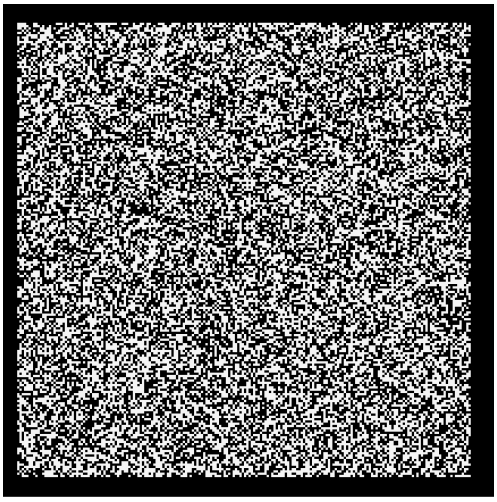
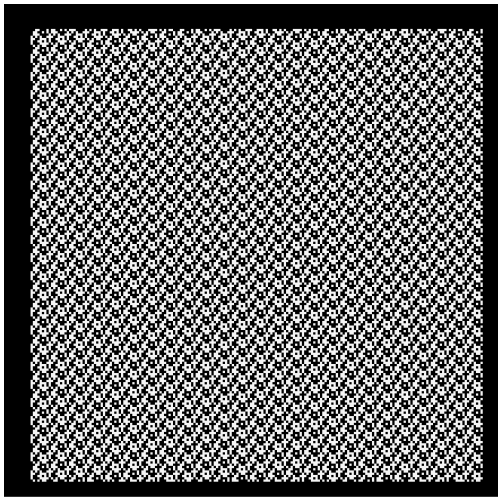
Random seed → different results each run.

```
#include <random>

int main()
{
  std::ranlux24_base rng(std::random_device{}()); // random seed
  std::uniform_int_distribution<int> dist(1, 6); // [1..6]

  int roll = dist(rng);
  return 0;
}
```

## PRNG QUALITY COMPARISON EXAMPLE

Visual

**Higher quality:** uniform distribution, no visible patterns.



**Lower quality:** clustering and structure.